

ANALYSIS AND VISUALIZATION OF CUSTOMER SHOPPING DATASET IN THE UNITED STATES

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1. DATASET

1.1 DESCRIPTION OF DATASET

The dataset shows the documentation on how people in the United States shop. It has information on 3,900 shopping trips (3,900 rows and 19 columns), with details like age, gender, what was bought, how much it cost, and customer reviews. We can use this data to understand how people shop, what they like, and how their shopping habits change over time.

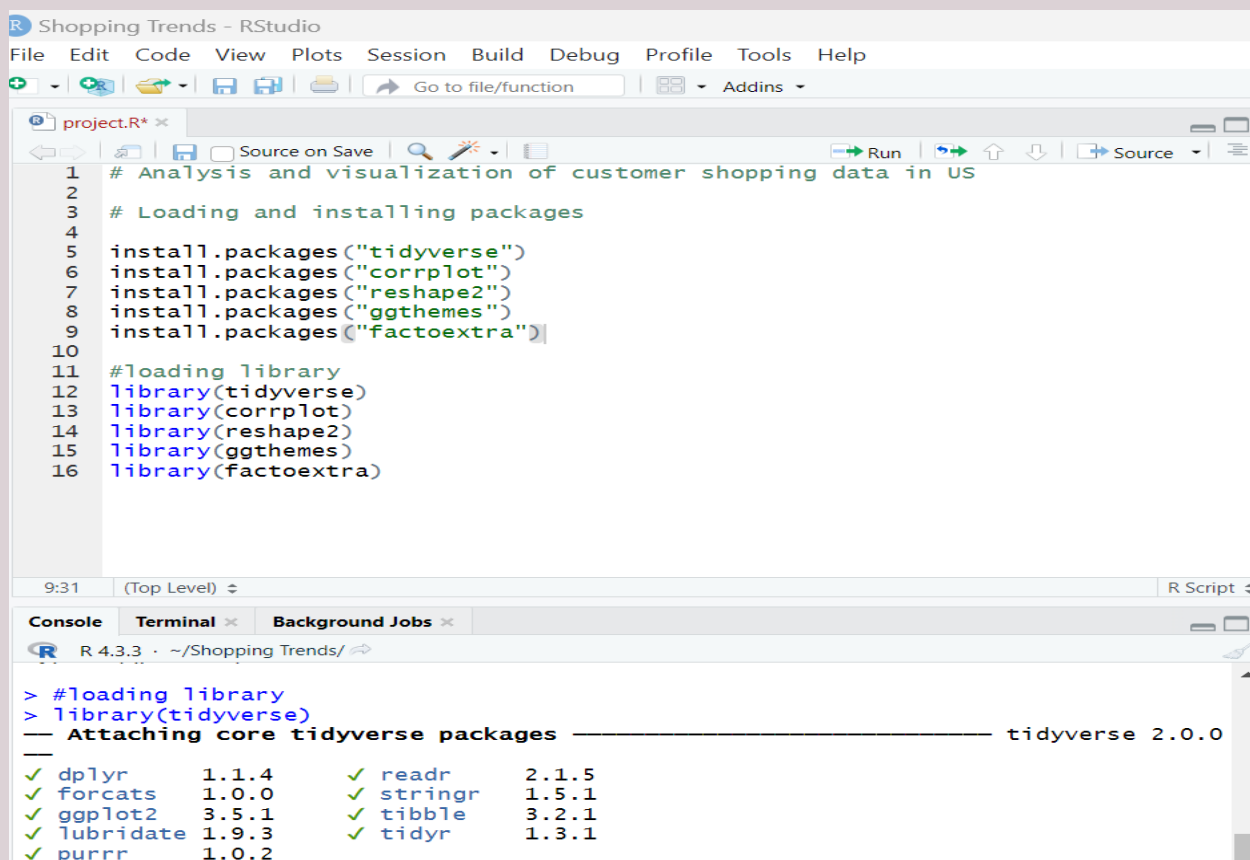
This data was gotten from Kaggle. It is called the "Customer Shopping Latest Trends Dataset" and can be viewed through this link:

<https://www.kaggle.com/datasets/bhadramohit/customer-shopping-latest-trends-dataset>

The dataset is about shopping in the United States. However, it does not say exactly when the data was collected. It is recent, but I cannot be sure.

1.2 INSTALLING PACKAGES AND LIBRARIES

Before I start my project, I get ready by installing packages that will help me understand and show the data. I installed packages like tidyverse and some other special tools. You can see some of the tools I installed in the picture below.



The screenshot shows the RStudio interface. The script editor displays the following R code:

```
1 # Analysis and visualization of customer shopping data in US
2
3 # Loading and installing packages
4
5 install.packages("tidyverse")
6 install.packages("corrplot")
7 install.packages("reshape2")
8 install.packages("ggthemes")
9 install.packages("factoextra")
10
11 #loading library
12 library(tidyverse)
13 library(corrplot)
14 library(reshape2)
15 library(ggthemes)
16 library(factoextra)
```

The console output shows the successful installation and loading of the tidyverse package and its core components:

```
> #loading library
> library(tidyverse)
── Attaching core tidyverse packages ─────────── tidyverse 2.0.0 ──
✓ dplyr      1.1.4      ✓ readr      2.1.5
✓ forcats    1.0.0      ✓ stringr   1.5.1
✓ ggplot2     3.5.1      ✓ tibble    3.2.1
✓ lubridate   1.9.3      ✓ tidyr     1.3.1
✓ purrr       1.0.2
```

Figure 1

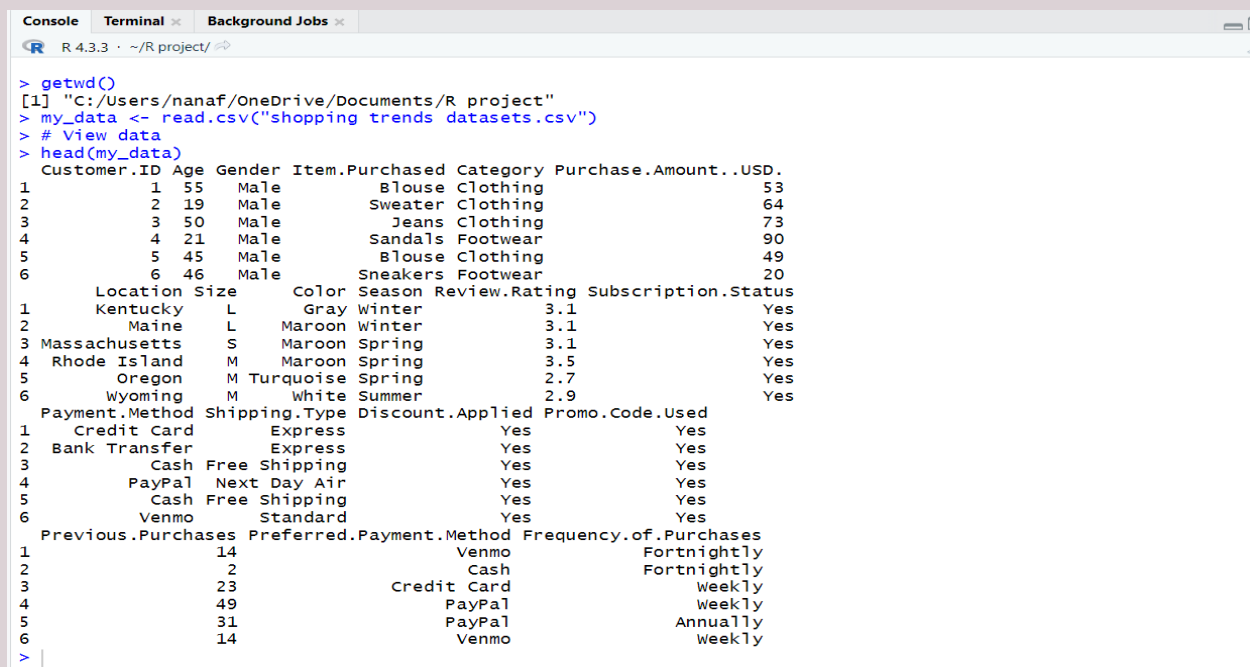
1.3 READING THE .CSV FILE

While I was adding my data to R, I used a function called `getwd()` to see where my files were saved on my computer.

```
> getwd()
[1] "C:/Users/nanaf/OneDrive/Documents/R project"
```

I used the `head()` function to view the first few rows (typically 6) of the data (see Figure 2). This helped me confirm that all the fields were imported successfully and appeared as expected in R.

I saved the data file (a .csv file) in my Documents folder. This way, it's easy to find when I'm working in R. Then, I used RStudio to open the data file. Here's how I did it:



```
Console Terminal Background Jobs
R 4.3.3 ~ /R project/

> getwd()
[1] "C:/Users/nanaf/OneDrive/Documents/R project"
> my_data <- read.csv("shopping trends datasets.csv")
> # View data
> head(my_data)
  Customer.ID Age Gender Item.Purchased Category Purchase.Amount..USD.
1           1  55   Male         Blouse Clothing                53
2           2  19   Male         Sweater Clothing                64
3           3  50   Male          Jeans Clothing                73
4           4  21   Male        Sandals Footwear                90
5           5  45   Male         Blouse Clothing                49
6           6  46   Male        Sneakers Footwear                20
  Location Size Color Season Review.Rating Subscription.Status
1 Kentucky L Gray Winter 3.1 Yes
2 Maine L Maroon Winter 3.1 Yes
3 Massachusetts S Maroon Spring 3.1 Yes
4 Rhode Island M Maroon Spring 3.5 Yes
5 Oregon M Turquoise Spring 2.7 Yes
6 Wyoming M White Summer 2.9 Yes
  Payment.Method Shipping.Type Discount.Applied Promo.Code.Used
1 Credit Card Express Yes Yes
2 Bank Transfer Express Yes Yes
3 Cash Free Shipping Yes Yes
4 PayPal Next Day Air Yes Yes
5 Cash Free Shipping Yes Yes
6 Venmo Standard Yes Yes
  Previous.Purchases Preferred.Payment.Method Frequency.of.Purchases
1 14 Venmo Fortnightly
2 2 Cash Fortnightly
3 23 Credit Card weekly
4 49 PayPal weekly
5 31 PayPal Annually
6 14 Venmo weekly
>
```

Figure 2

- I used the `is.data.frame()` function to verify that the data loaded as a data frame, the preferred format for working with data in R.
- Next, I used `dim(my_data)` to check the dimensions of the data, which tells me the number of rows (observations) and columns (variables).

```

> dim(my_data)
[1] 3900 19
> is.data.frame(my_data)
[1] TRUE
> |

```

Figure 3

The `str()` function was used to examine the structure of the data, including the data types of each variable. This helped identify any variables that might need data type conversions or other adjustments.

```

Console Terminal Background Jobs
R 4.3.3 ~ /R project/
> str(my_data)
'data.frame':   3900 obs. of  19 variables:
 $ Customer.ID      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Age              : int  55 19 50 21 45 46 63 27 26 57 ...
 $ Gender           : chr  "Male" "Male" "Male" "Male" ...
 $ Item.Purchased   : chr  "Blouse" "Sweater" "Jeans" "Sandals" ...
 $ Category         : chr  "Clothing" "Clothing" "Clothing" "Footwear" ...
 $ Purchase.Amount..USD.: int  53 64 73 90 49 20 85 34 97 31 ...
 $ Location         : chr  "Kentucky" "Maine" "Massachusetts" "Rhode Island" ...
 $ Size            : chr  "L" "L" "S" "M" ...
 $ Color           : chr  "Gray" "Maroon" "Maroon" "Maroon" ...
 $ Season          : chr  "Winter" "Winter" "Spring" "Spring" ...
 $ Review.Rating    : num  3.1 3.1 3.1 3.5 2.7 2.9 3.2 3.2 2.6 4.8 ...
 $ Subscription.Status: chr  "Yes" "Yes" "Yes" "Yes" ...
 $ Payment.Method   : chr  "Credit Card" "Bank Transfer" "Cash" "PayPal" ...
 $ Shipping.Type    : chr  "Express" "Express" "Free Shipping" "Next Day Air" ...
 $ Discount.Applied : chr  "Yes" "Yes" "Yes" "Yes" ...
 $ Promo.Code.Used  : chr  "Yes" "Yes" "Yes" "Yes" ...
 $ Previous.Purchases: int  14 2 23 49 31 14 49 19 8 4 ...
 $ Preferred.Payment.Method: chr  "Venmo" "Cash" "Credit Card" "PayPal" ...
 $ Frequency.of.Purchases: chr  "Fortnightly" "Fortnightly" "Weekly" "Weekly" ...
> |

```

Figure 4

The `colnames()` function was used to list the names of all 19 variables in the dataset.

```

> colnames(my_data)
[1] "Customer.ID"      "Age"      "Gender"
[4] "Item.Purchased"   "Category"  "Purchase.Amount..USD."
[7] "Location"         "Size"     "Color"
[10] "Season"          "Review.Rating" "Subscription.Status"
[13] "Payment.Method"  "Shipping.Type" "Discount.Applied"
[16] "Promo.Code.Used" "Previous.Purchases" "Preferred.Payment.Method"
[19] "Frequency.of.Purchases"
> |

```

Figure 5

1.4 TABLE OF VARIABLES AND DESCRIPTION

Variable Name	Mode	Description
Customer ID	int	Unique identifier for each customer.
Age	int	The age of the customer in years.
Gender	chr	The gender of the customer (e.g., Male, Female).
Item Purchased	chr	Name of the item purchased by the customer.
Category	chr	The category of the purchased item (e.g., Clothing, Footwear).
Purchase Amount (USD)	int	The total amount spent by the customer in USD.
Location	chr	The geographic location of the customer.
Size	chr	The size of the purchased item (e.g., S, M, L).
Color	chr	The color of the purchased item.

Season	chr	The season in which the purchase was made (e.g., Spring, Winter).
Review Rating	num	The customer's rating for the product, ranging from 1 to 5.
Subscription Status	chr	Indicates whether the customer is subscribed (e.g., Yes, No).
Payment Method	chr	The payment method used (e.g., Credit Card, PayPal).
Shipping Type	chr	The shipping method selected (e.g., Free Shipping, Express).
Discount Applied	chr	Indicates if a discount was applied (e.g., Yes, No).
Promo Code Used	chr	Indicates if a promotional code was used (e.g., Yes, No).
Previous Purchases	int	The number of prior purchases made by the customer.
Preferred Payment Method	chr	The customer's preferred payment method.
Frequency of Purchases	chr	How often the customer makes purchases (e.g., Weekly, Monthly, Annually).

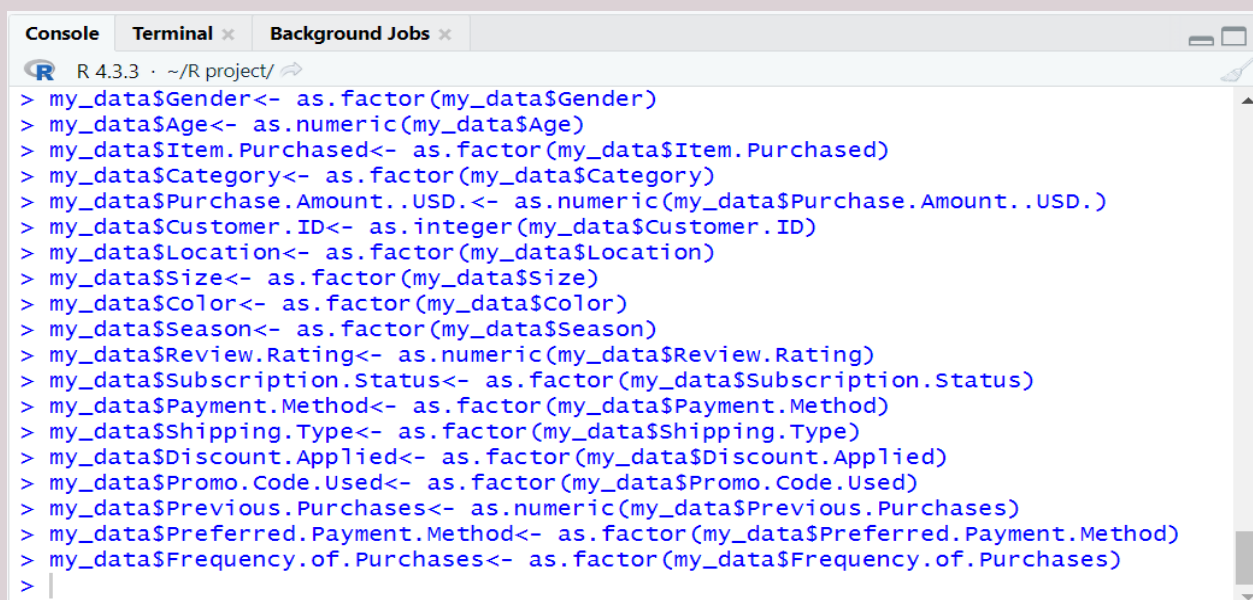
1.5 EXPECTATION

I expect this analysis to reveal patterns in how customers shop, like the most popular product categories, seasonal trends, and preferences based on demographics. I also aim to find connections between factors such as review ratings, how much customers spend, and how often they make purchases.

1.6 DATA CLEANING

I checked the data to see if anything was missing. Everything appeared to be present, and there were no duplicates. However, the variables were not properly defined. To address this, I cleaned the data to make it clearer and easier to understand and analyze.

I changed the way some of the data were categorized. For example, I changed from “chr” to “Factor”; from “int” to “Factor”; from “int” to “num”, etc.)



```

R 4.3.3 · ~/R project/
> my_data$Gender<- as.factor(my_data$Gender)
> my_data$Age<- as.numeric(my_data$Age)
> my_data$Item.Purchased<- as.factor(my_data$Item.Purchased)
> my_data$Category<- as.factor(my_data$Category)
> my_data$Purchase.Amount..USD.<- as.numeric(my_data$Purchase.Amount..USD.)
> my_data$Customer.ID<- as.integer(my_data$Customer.ID)
> my_data$Location<- as.factor(my_data$Location)
> my_data$Size<- as.factor(my_data$Size)
> my_data$Color<- as.factor(my_data$Color)
> my_data$Season<- as.factor(my_data$Season)
> my_data$Review.Rating<- as.numeric(my_data$Review.Rating)
> my_data$Subscription.Status<- as.factor(my_data$Subscription.Status)
> my_data$Payment.Method<- as.factor(my_data$Payment.Method)
> my_data$Shipping.Type<- as.factor(my_data$Shipping.Type)
> my_data$Discount.Applied<- as.factor(my_data$Discount.Applied)
> my_data$Promo.Code.Used<- as.factor(my_data$Promo.Code.Used)
> my_data$Previous.Purchases<- as.numeric(my_data$Previous.Purchases)
> my_data$Preferred.Payment.Method<- as.factor(my_data$Preferred.Payment.Method)
> my_data$Frequency.of.Purchases<- as.factor(my_data$Frequency.of.Purchases)
>

```

Figure 6

The cleaned data now includes 4 numerical variables and 15 categorical variables and the “str function” was used to show the new data, as detailed below:

```

Console Terminal Background Jobs
R 4.3.3 · ~/R project/
> str(my_data)
'data.frame': 3900 obs. of 19 variables:
 $ Customer.ID      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Age              : num  38 2 33 4 28 29 46 10 9 40 ...
 $ Gender           : Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 2 2 2 2 ...
 $ Item.Purchased   : Factor w/ 25 levels "Backpack","Belt",...: 3 24 12 15 3 21 17 19 5 8 ...
 $ Category         : Factor w/ 4 levels "Accessories",...: 2 2 2 3 2 3 2 2 4 1 ...
 $ Purchase.Amount..USD.: num  53 64 73 90 49 20 85 34 97 31 ...
 $ Location         : Factor w/ 50 levels "Alabama","Alaska",...: 17 19 21 39 37 50 26 18 48 25 ...
 $ Size             : Factor w/ 4 levels "L","M","S","XL": 1 1 3 2 2 2 2 1 1 2 ...
 $ Color            : Factor w/ 25 levels "Beige","Black",...: 8 13 13 13 22 24 8 5 20 17 ...
 $ Season           : Factor w/ 4 levels "Fall","Spring",...: 4 4 2 2 2 3 1 4 3 2 ...
 $ Review.Rating    : num  3.1 3.1 3.1 3.5 2.7 2.9 3.2 3.2 2.6 4.8 ...
 $ Subscription.Status : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2 2 2 2 ...
 $ Payment.Method    : Factor w/ 6 levels "Bank Transfer",...: 3 1 2 5 2 6 4 4 6 5 ...
 $ Shipping.Type     : Factor w/ 6 levels "2-Day Shipping",...: 2 2 3 4 3 5 3 3 2 1 ...
 $ Discount.Applied  : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2 2 2 2 ...
 $ Promo.Code.Used   : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2 2 2 2 ...
 $ Previous.Purchases : num  14 2 23 49 31 14 49 19 8 4 ...
 $ Preferred.Payment.Method: Factor w/ 6 levels "Bank Transfer",...: 6 2 3 5 5 6 2 3 6 2 ...
 $ Frequency.of.Purchases : Factor w/ 7 levels "Annually","Bi-Weekly",...: 4 4 7 7 1 7 6 7 1 6 ...

```

Figure 7

Here's the output from the head () function and it did not change

```

Console Terminal Background Jobs
R 4.3.3 · ~/R project/
> head(my_data)
  Customer.ID Age Gender Item.Purchased Category Purchase.Amount..USD. Location Size Color Season
1          1  38   Male      Blouse Clothing             53      Kentucky  L    Gray  Winter
2          2   2   Male      Sweater Clothing             64         Maine  L   Maroon  Winter
3          3  33   Male       Jeans Clothing             73 Massachusetts S   Maroon  Spring
4          4   4   Male     Sandals Footwear             90      Rhode Island M   Maroon  Spring
5          5  28   Male      Blouse Clothing             49         Oregon  M Turquoise Spring
6          6  29   Male     Sneakers Footwear             20        Wyoming  M    White  Summer
  Review.Rating Subscription.Status Payment.Method Shipping.Type Discount.Applied Promo.Code.Used
1          3.1              Yes      Credit Card      Express             Yes             Yes
2          3.1              Yes      Bank Transfer      Express             Yes             Yes
3          3.1              Yes           Cash Free Shipping             Yes             Yes
4          3.5              Yes       PayPal Next Day Air             Yes             Yes
5          2.7              Yes           Cash Free Shipping             Yes             Yes
6          2.9              Yes           Venmo Standard             Yes             Yes
  Previous.Purchases Preferred.Payment.Method Frequency.of.Purchases
1             14          Venmo Fortnightly
2              2             Cash Fortnightly
3             23       Credit Card      weekly
4             49        PayPal      weekly
5             31        PayPal      Annually
6             14          Venmo      weekly

```

Figure 8

Table of Variables and Descriptions After cleaning

Variable Name	Mode	Description
Customer ID	int	Unique identifier for each customer.
Age	num	The age of the customer in years.
Gender	fac	The gender of the customer (e.g., Male, Female).
Item Purchased	fac	Name of the item purchased by the customer.
Category	fac	The category of the purchased items (e.g., Clothing, Footwear).

Purchase Amount (USD)	num	The total amount spent by the customer in USD.
Location	fac	The geographic location of the customer.
Size	fac	The size of the purchased item (e.g., S, M, L).
Color	fac	The color of the purchased item.
Season	fac	The season in which the purchase was made (e.g., Spring, Winter).
Review Rating	num	The customer's rating for the product, ranging from 1 to 5.
Subscription Status	fac	Indicates whether the customer is subscribed (e.g., Yes, No).
Payment Method	fac	The payment method used (e.g., Credit Card, PayPal).
Shipping Type	fac	The shipping method selected (e.g., Free Shipping, Express).
Discount Applied	fac	Indicates if a discount was applied (e.g., Yes, No).
Promo Code Used	fac	Indicates if a promotional code was used (e.g., Yes, No).
Previous Purchases	num	The number of prior purchases made by the customer.
Preferred Payment Method	fac	The customer's preferred payment method.
Frequency of Purchases	fac	How often the customer makes purchases (e.g., Weekly, Monthly, Annually).

2. DATA ANALYSIS

2.1 SCATTERPLOTS AND CORRELATION MATRICES OF NUMERIC VARIABLES

The dataset includes different types of variables, such as numeric and categorical variables. I focused on the numerical variables to find patterns and connections between them. To do this, I used the `cor()` function, which helps check how much the variables are related to one another.

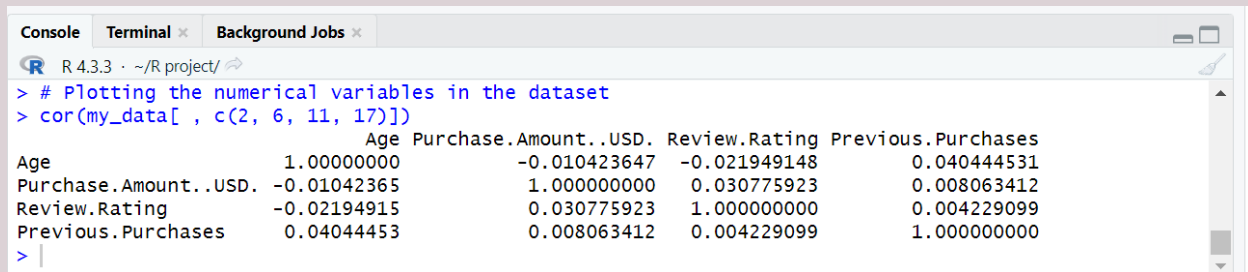


Figure 9

The correlation table reveals that the variables in the dataset have very weak relationships with each other. Age, Purchase Amount, Review Rating, and Previous Purchases are largely independent, showing almost no linear connection. This suggests that these variables don't significantly influence one another within the dataset.

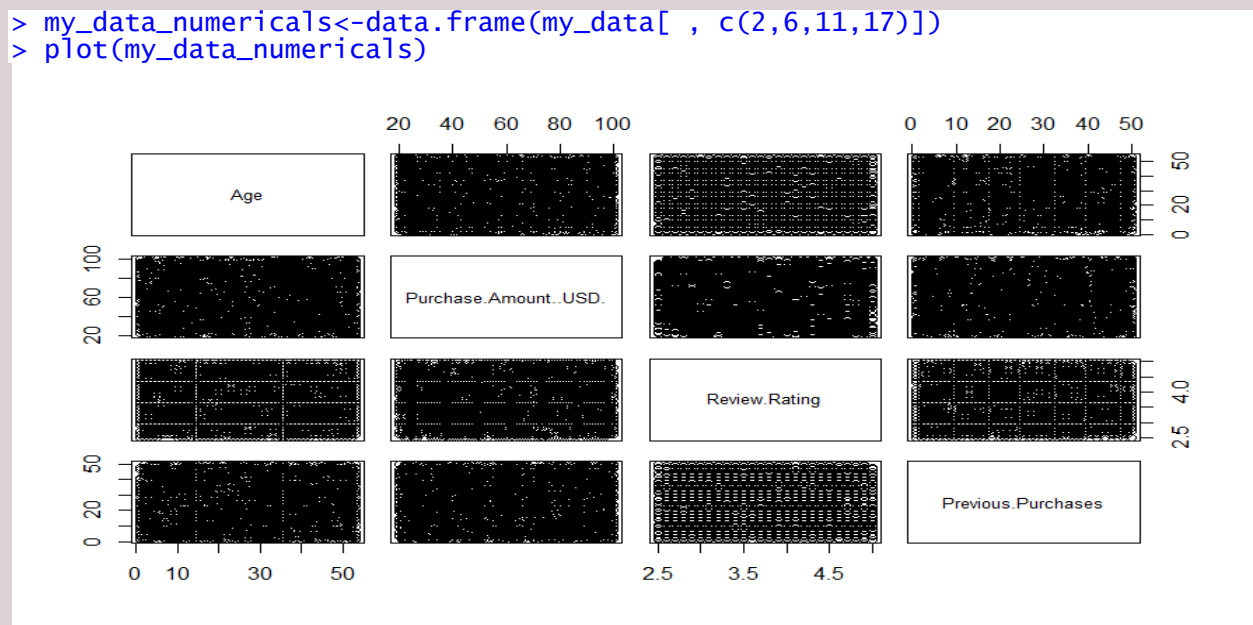


Figure 10. Scatterplot Matrix

Corrplot was also used to visualize the correlation as shown below

```
corrplot(cor(my_data_numericals), type= "full", tl.col="navy", bg="white", col=1=NULL )
```

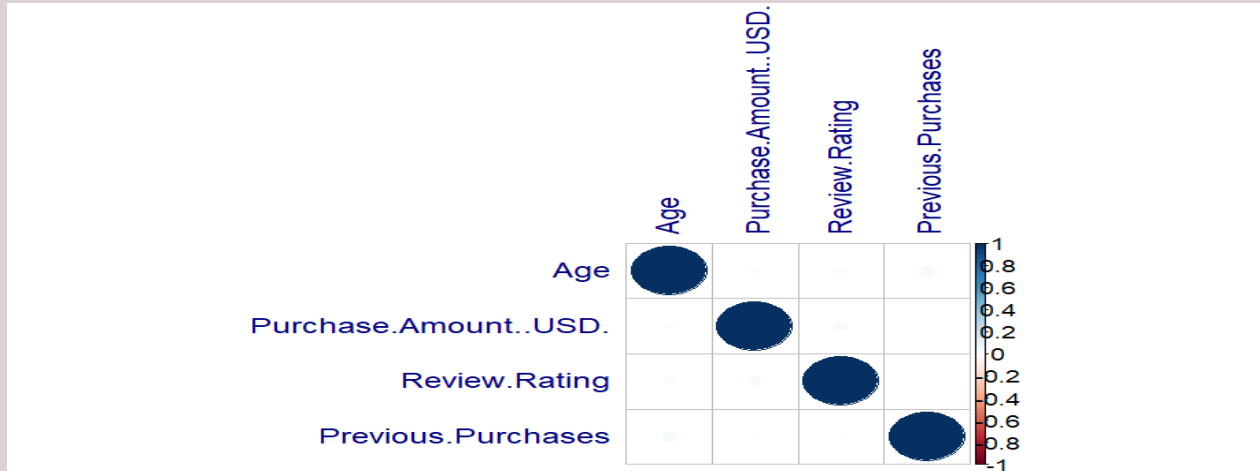


Figure 11. Corrplot

2.1.1 CORRELATION BETWEEN PURCHASE AMOUNT (USD) AND REVIEW RATING

I created a scatter plot to visualize the relationship between Purchase Amount and Review Rating. While there's a slight positive trend, it's not a strong one. This suggests a weak positive correlation between the two variables.

```
> plot(my_data$Purchase.Amount..USD., my_data$Review.Rating, main ="Purchase Amount and Review Rating", xlab="Purchase amount", ylab="Review Rating")
```

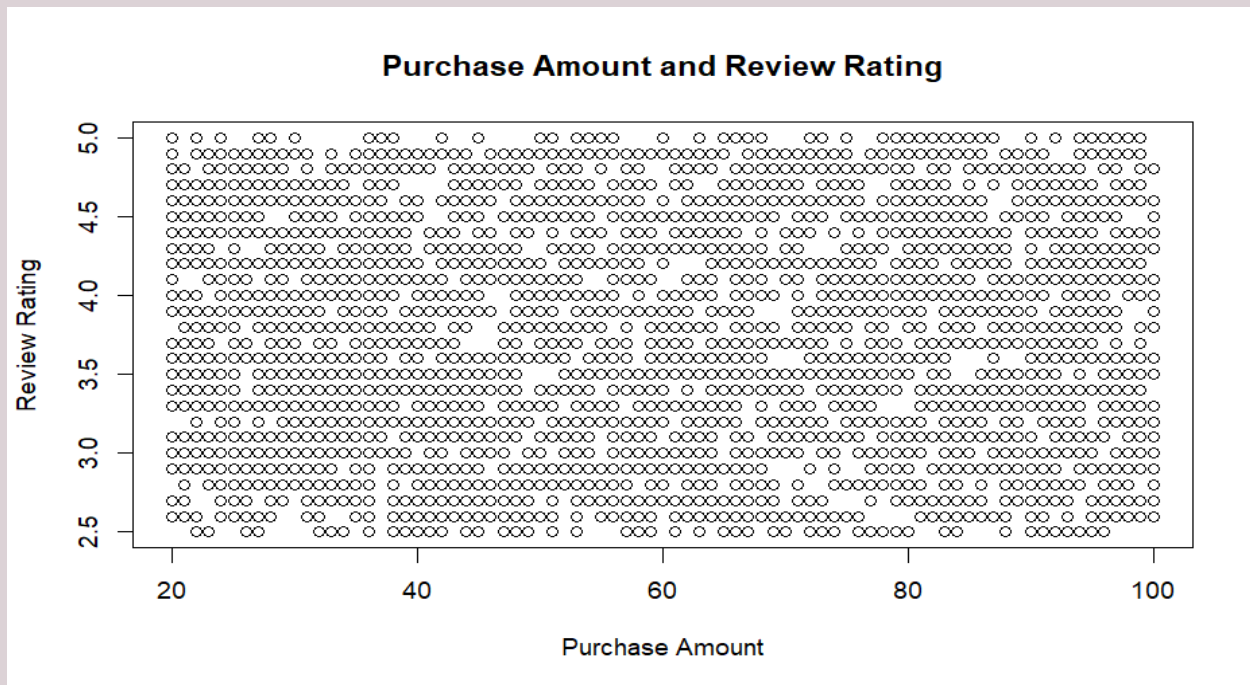


Figure 12. Scatter Plot on Purchase Amount and Review Rating

2.1.2. CORRELATION BETWEEN REVIEW RATING AND PREVIOUS PURCHASES

The scatter plot shows a weak positive link between Previous Purchases and Review Rating. As previous purchases increase, review ratings go up slightly, but the connection is not strong, and the points are very spread out.

```
> plot(my_data$Review.Rating, my_data$Previous.Purchases, main = "Review Rating and Previous Purchase", xlab = "Review Rating", ylab = "Previous Purchase")
```

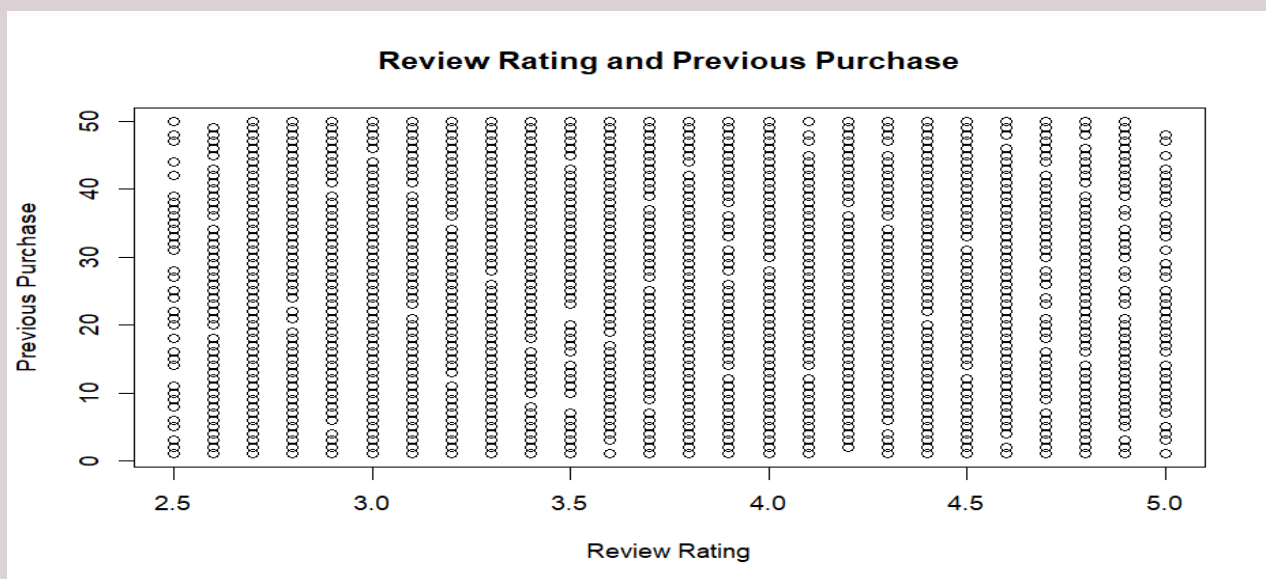


Figure 13 Scatter Plot on Review Rating and Previous Purchase

2.1.3 CORRELATION BETWEEN PREVIOUS PURCHASES AND PURCHASE AMOUNT (USD)

The scatter plot shows a weak positive relationship between Previous Purchases and Purchase Amount. As previous purchases increase, the purchase amount tends to rise slightly. However, the connection is not strong, and the data points are widely scattered.

```
> plot(my_data$Previous.Purchases, my_data$Purchase.Amount..USD., main = "Previous Purchase and Purchase Amount", xlab = "Previous Purchase", ylab = "Purchase Amount")
```

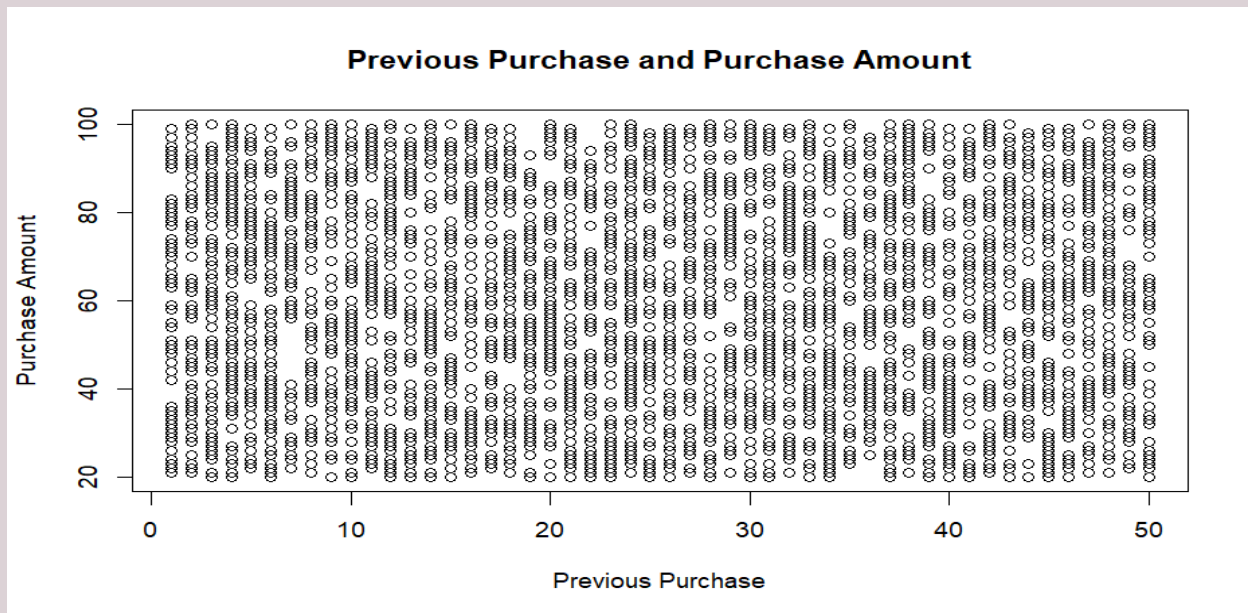


Figure 14 Scatter Plot on Previous Purchase and Purchase Amount

2.2 SUMMARY STATISTICS

The `summary()` function in R was very useful for quickly analyzing the dataset. I excluded only the Customer.ID column and found some interesting patterns. It provided insights into customer categories, purchase frequency, sizes, gender, and more. These summaries would be helpful for visualizing trends and understanding customer segments better.

```
> summary(my_data[,c(-1)])
```

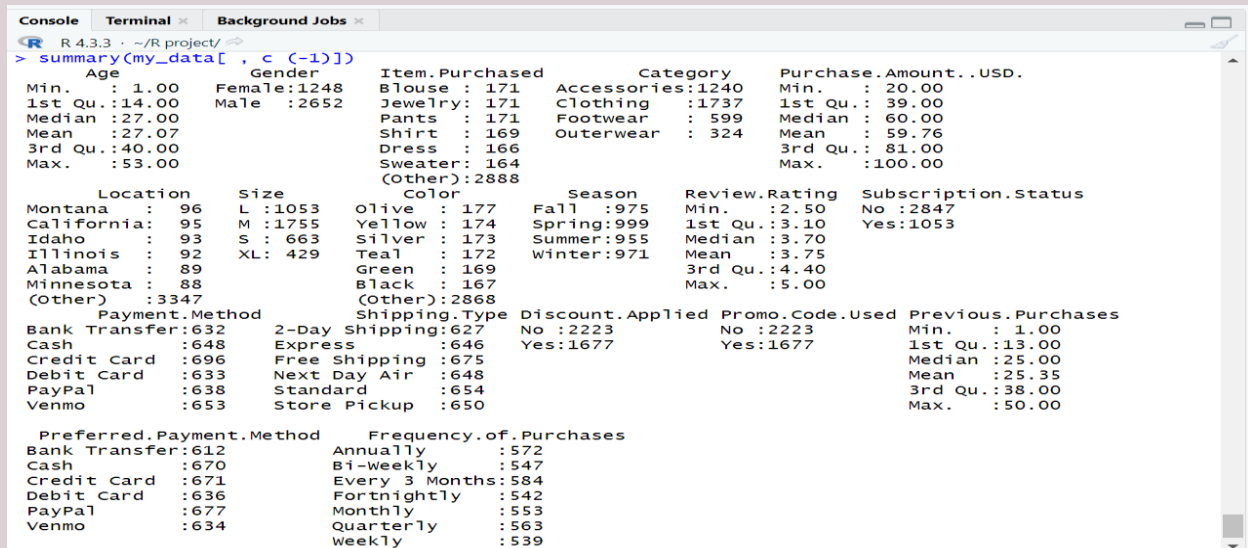


Figure 15. Summary Statistic Data

Next, I calculated summary statistics for the numerical variables. These statistics provide key insights into the distribution and central tendency of numerical data.

Summary Statistics of numerical variables.

```
> summary(my_data_numericals)
```

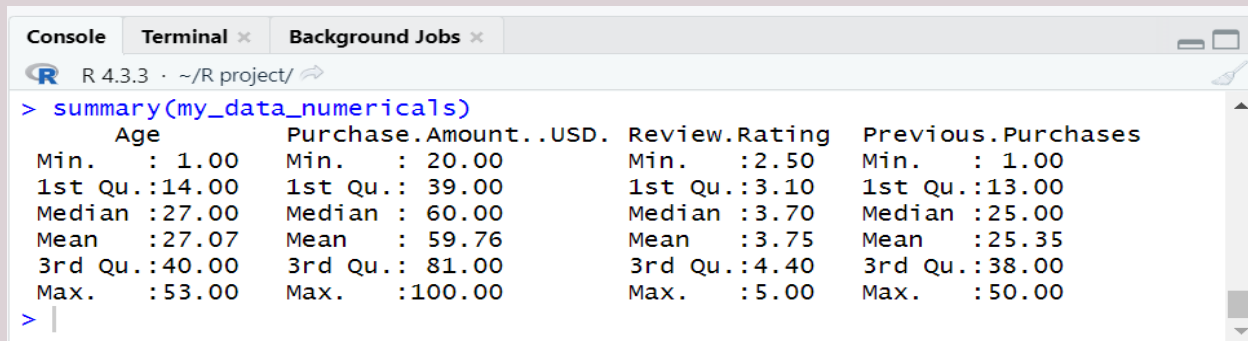


Figure 16.

The summary displays the mean and median values, along with other statistical measures, for the numerical variables.

2.3 BAR PLOTS

In Figure 15 above, visualizing gender in a bar plot would be helpful as it can show which gender shops more.

I utilized the `table()` function to extract the \$Gender column, which I then used to generate the bar plot.

```
> table(my_data$Gender)
```

```

Female    Male
 1248     2652

> #Percentage Distribution:
> 1248/3900 #Female
[1] 0.32
> 2652/3900 #Male
[1] 0.68
>
> barplot(table(my_data$Gender), main = "Customer Shopping by Gender", xlab =
"Gender", ylab = "No. of Purchases")

```

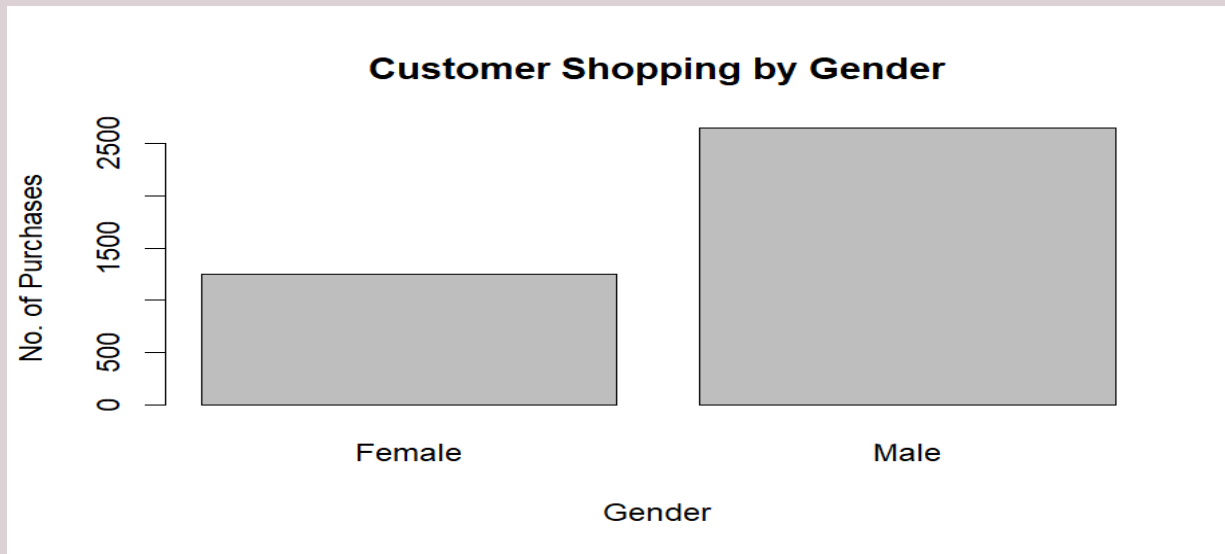


Figure 17. Barplot of Customers Shopping Base on Gender

(The bar chart shows the distribution of purchases by gender. Males seem to make significantly more purchases than females, suggesting a potential gender disparity in shopping behavior.)

I also created a bar plot for the product categories to provide more details about the types of products purchased.

```

> table(my_data$Category)

Accessories    Clothing    Footwear    Outerwear
      1240         1737         599         324

>
> #Percentage Distribution:
> 1240/3900 #Accessories
[1] 0.3179487
> 1737/3900 #Clothing
[1] 0.4453846
> 599/3900 #Footwear
[1] 0.1535897
> 324/3900 #Outerwear
[1] 0.08307692
>
> barplot(table(my_data$Category), main = "Products Shopped by Customers", xlab =
"Category", ylab = "No. of Purchases")

```



Figure 18. Barplot of What Customers Shopped For

(The plot above shows the distribution of purchases across various product categories. Clothing stands out as the most popular category, followed by accessories, footwear, and outerwear. This indicates that customers have a strong preference for clothing over the other categories.)

Another interesting visualization is the frequency of purchases, as customers are consistently making purchases.

```
> table(my_data$Frequency.of.Purchases)
      Annually      Bi-weekly Every 3 Months      Fortnightly
      572        547        584        542
      Monthly      Quarterly      weekly
      553        563        539
>
> #Percentage Distribution:
> 572/3900 #Annually
[1] 0.1466667
> 553/3900 #Monthly
[1] 0.1417949
> 547/3900 #Bi-weekly
[1] 0.1402564
> 563/3900 #Quarterly
[1] 0.144359
> 584/3900 #Every 3 Months
[1] 0.1497436
> 539/3900 #Weekly
[1] 0.1382051
> 542/3900 #Fortnightly
[1] 0.1389744
>
> barplot(table(my_data$Frequency.of.Purchases), main = "Shopping Base on Ret
urning Customers", xlab = "Frequency of Purchases", ylab = "No. of Purchases"
)
```

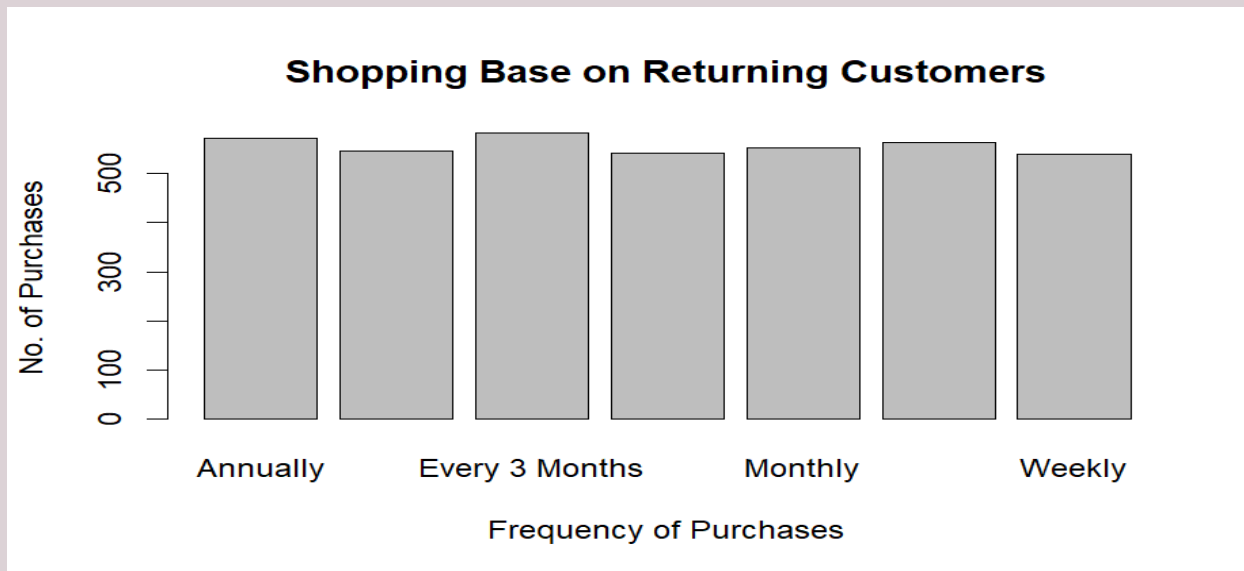


Figure 19. Barplot Showing Shopping Trends Base on Returning Customers

(The bar chart shows the distribution of purchases based on customer return frequency. Customers who shop every three months have the highest frequency, followed by annual and weekly shoppers. Monthly shoppers have the lowest frequency, suggesting that less frequent but consistent shoppers contribute more to overall sales.)

2.4 PIE CHART

Product sizes and the seasons when customers shop the most are important data points to explore and visualize. I used the `pie ()` function in R to create this visualization. It helps business owners understand shopping patterns and develop strategies accordingly.

Pie chart on Size

```
> table(my_data$Size)
  L    M    S   XL
1053 1755 663 429
>
> #Percentage Distribution:
> 1053/3900 #L
[1] 0.27
> 1755/3900 #M
[1] 0.45
> 663/3900 #S
[1] 0.17
> 429/3900 #XL
[1] 0.11
>
> pie(table(my_data$Size), main = "Customer Size")
```

Customer Size

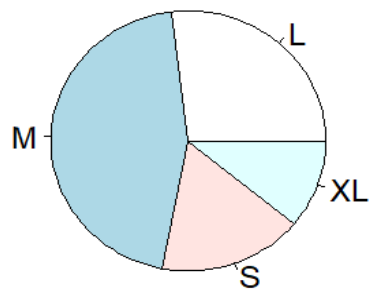


Figure 20. Pie Chart Showing Sizes Purchased the Most

(The pie chart shows that most customers prefer medium-sized clothing, followed by large and small sizes. Extra-large sizes are the least popular.)

Pie chart on Season

```
> table(my_data$Season)
  Fall Spring Summer winter
   975   999   955   971
>
> #Percentage Distribution:
> 975/3900 #Fall
[1] 0.25
> 999/3900 #Spring
[1] 0.2561538
> 955/3900 #Summer
[1] 0.2448718
> 971/3900 #winter
[1] 0.2489744
>
> pie(table(my_data$Season), main = "Shopping Base on Season")
```

Shopping Base on Season

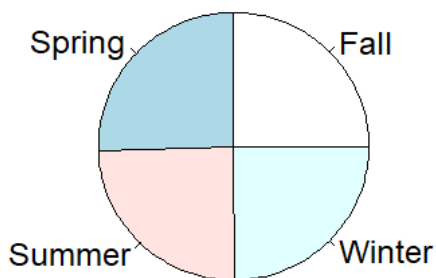


Figure 21. Pie Chart Showing Season Customers Shops the Most

(The pie chart shows that Fall and Winter are the most popular shopping seasons, followed by Spring and Summer, suggesting customers shop more during colder seasons.)

2.5 BOXPLOT

To identify outliers and inconsistencies between numerical and categorical variables, I used boxplots to examine discrepancies and explore the causes of the outliers.

The boxplots below display purchase amounts by customer gender, revealing that spending varies based on the shopper's gender.

```
my_data%>%  
+   ggplot(aes(Gender, Purchase.Amount..USD.)) +  
+   geom_boxplot()+  
+   theme_bw()+  
+   labs(x= "Gender", y="Purchase Amount")
```

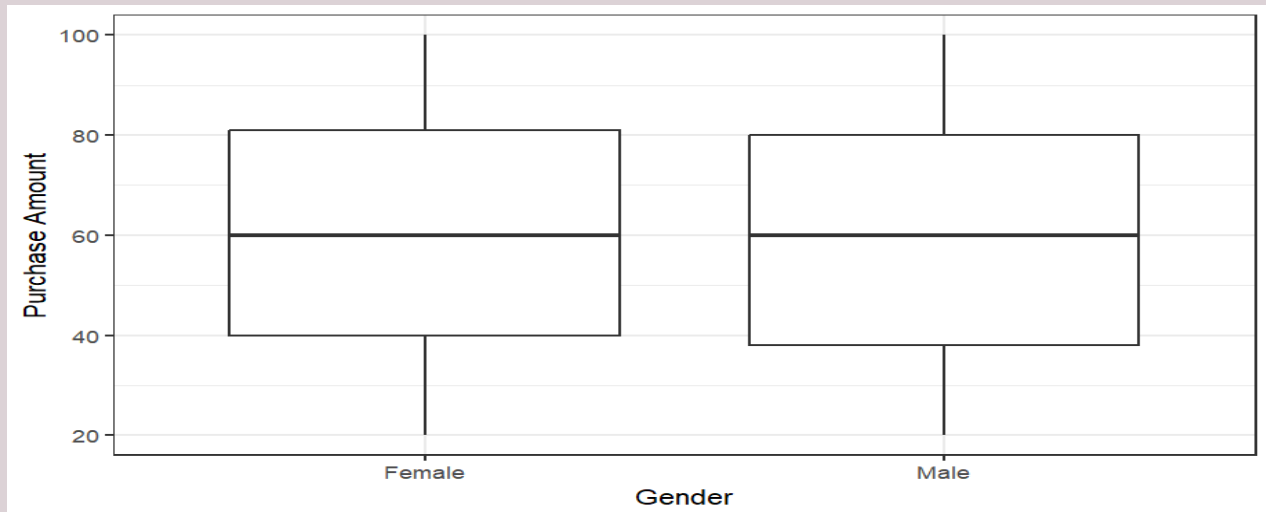


Figure 22. Boxplot on Gender and Purchase Amount

(The boxplot shows purchase amounts for females and males. Both have a similar median of about \$60, but males have a wider range with some very high and very low purchases. Females have a more consistent range with fewer extreme values.)

Another variable that showed variation when plotted against customer gender is previous purchases.

```
> my_data%>%
+ ggplot(aes(Gender, Previous.Purchases)) +
+ geom_boxplot()+
+ theme_bw()+
+ labs(x= "Gender", y="Previous Purchases")
```

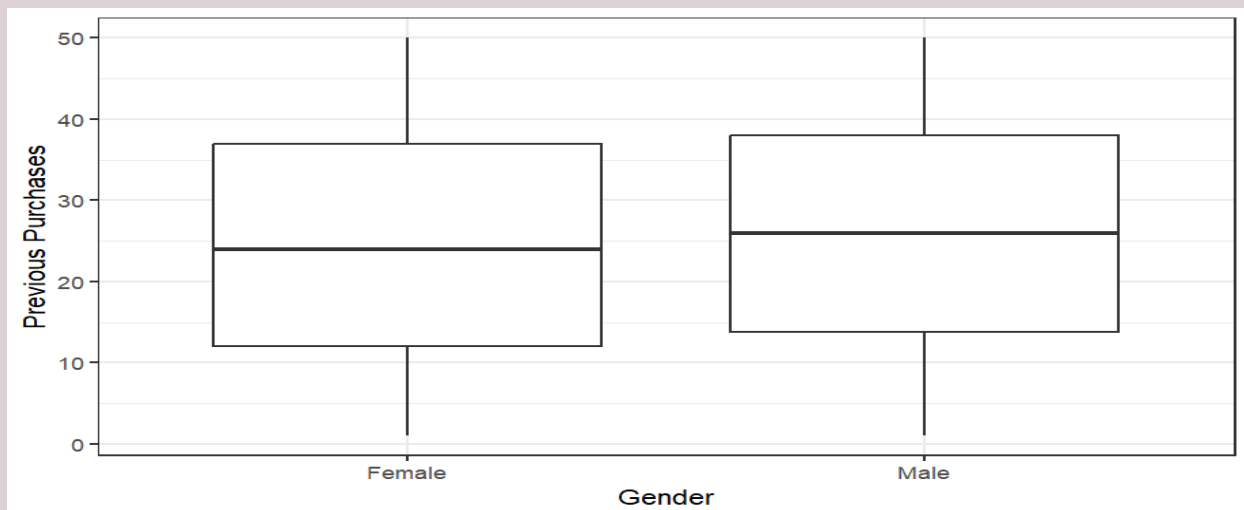


Figure 23. Boxplot on Gender and Previous Purchases

(The boxplot shows that both genders have a similar number of previous purchases on average. However, males have a wider range of purchases, with some extreme values on both ends. In

contrast, females have a more consistent range with fewer outliers. This indicates that while both genders have a similar average, males show more variation in their purchasing behavior than females.)

2.6 CLOSER LOOK AT PURCHASE AMOUNT

I noticed a slight positive trend between Purchase Amount and Review Rating, but the relationship isn't strong. The number of previous purchases doesn't seem to significantly impact either variable. This suggests that while higher-priced items may slightly correlate with higher ratings, the number of previous purchases doesn't strongly influence this connection.

The bubble chart shows the relationship between Purchase Amount, Review Rating, and the number of Previous Purchases, with the bubble size representing the number of previous purchases.

```
> my_data %>%  
+   ggplot(aes(Review.Rating, Purchase.Amount..USD.)) +  
+   geom_point(aes(size= Previous.Purchases))+  
+   coord_flip()+  
+   theme_classic()
```

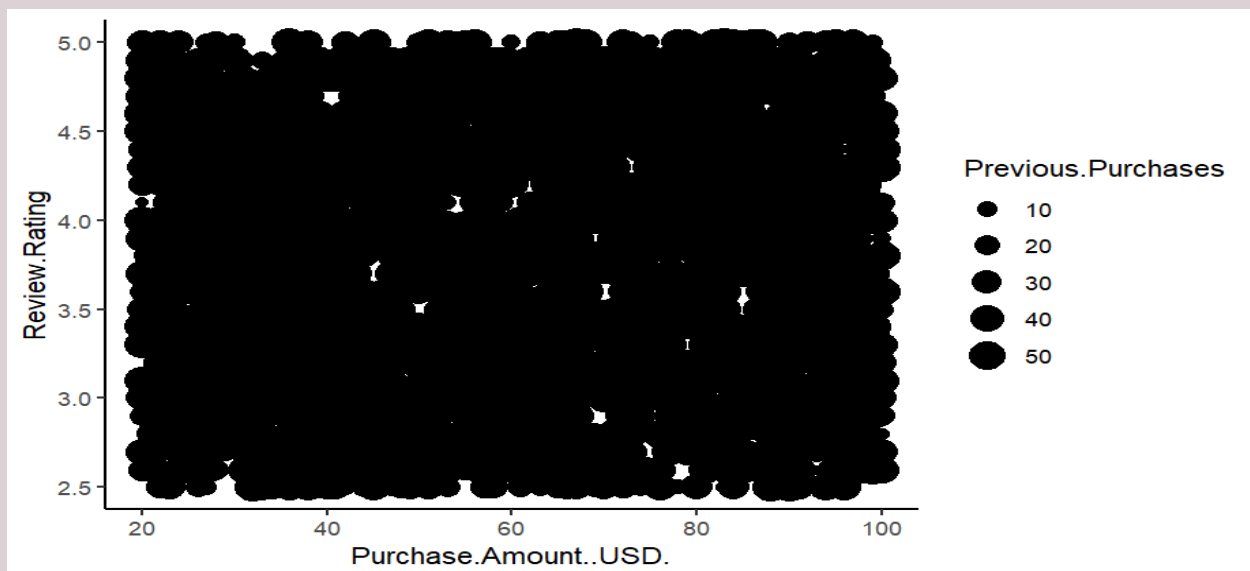


Figure 24. The Influence of Review Rating on Customer Behavior

From another notable analysis I did, there seems to be no notable difference in spending between states. The dot plot shows the distribution of purchase amounts across different states. The dots are evenly spread across the purchase amount range in each state, indicating that customers from various states generally spend similar amounts on average.

```
> my_data %>%  
+   ggplot(aes(Location, Purchase.Amount..USD.)) +  
+   geom_point()+  
+   coord_flip()+
```

```
+ theme_gray()
```



Figure 25 Customer Demographics on Purchase Patterns

Next, let's examine the product categories and identify which one was purchased the most. I created a density chart to compare the four categories.

```
> my_data %>%
+ ggplot(aes(Purchase.Amount..USD.)) +
+ geom_density()+
+ facet_wrap(~Category)
+ theme_bw()
```

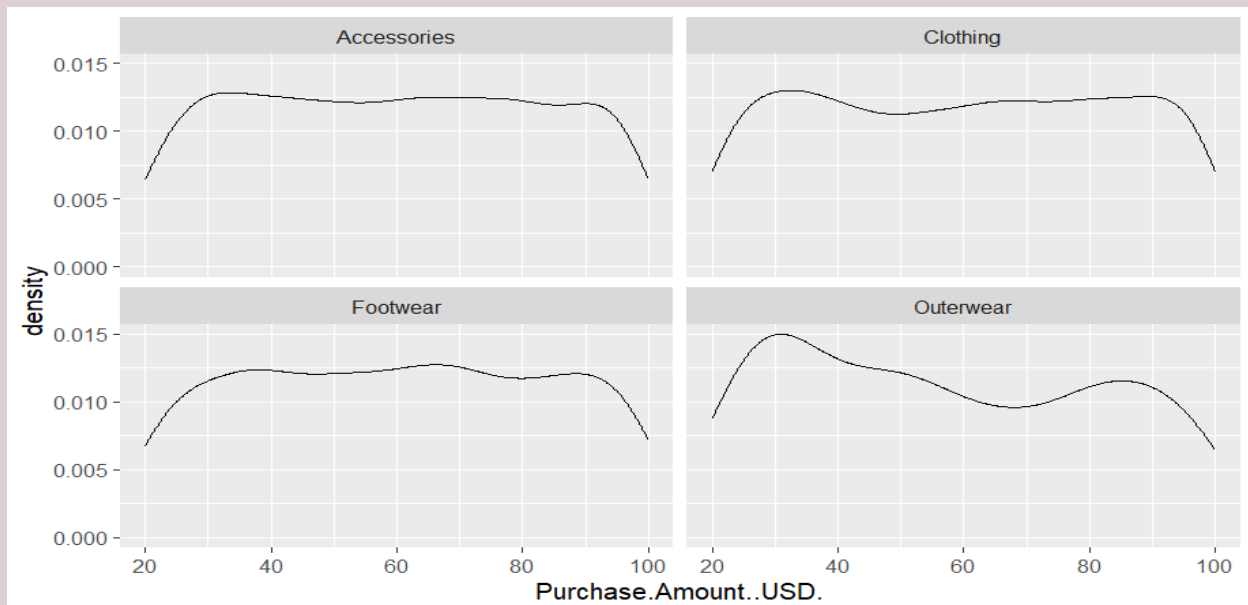


Figure 26. Purchase Per Category

(Accessories and Footwear have a wide range with no clear peak, while Clothing and Outerwear show lower-priced items, indicating a right-skewed distribution.)

2.7 HISTOGRAM AND DENSITY PLOT

I also analyzed the purchase behavior of all orders by customers, as this helps vendors track fast-moving products and develop effective strategies for better understanding their clients.

I plotted a histogram to show the number of orders placed at each purchase amount. The distribution appears relatively uniform, with no significant peaks or valleys. This suggests that orders are spread across a wide range of purchase amounts, with no specific amount being notably more or less popular.

```
my_data %>%  
+   ggplot(aes(Purchase.Amount..USD.))+  
+   geom_histogram(stat = "count")+  
+   labs(x= "Purchase Amount",  
+        y= "No. of Orders")+  
+   theme_bw()
```

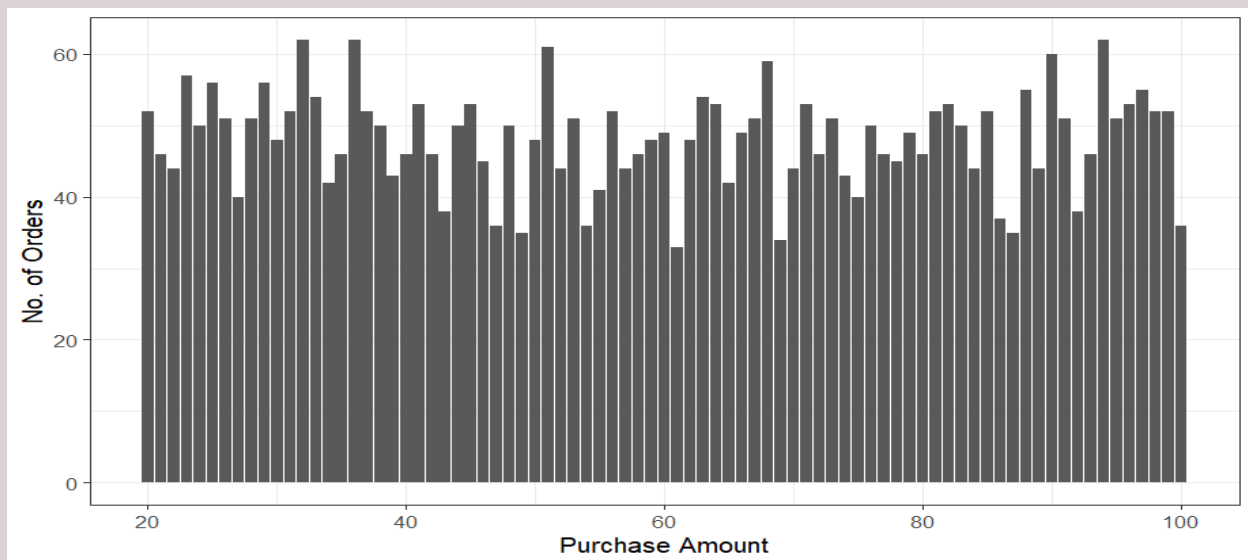


Figure 27. Purchase of Orders

This can also be represented in a density plot across various product categories. The results below highlight the items that customers most commonly purchase.

```
> my_data %>%  
+   ggplot(aes(Purchase.Amount..USD.))+  
+   geom_density(colour = "navyblue", fill = "seagreen")+  
+   facet_wrap(~Item.Purchased)+  
+   theme_classic()
```

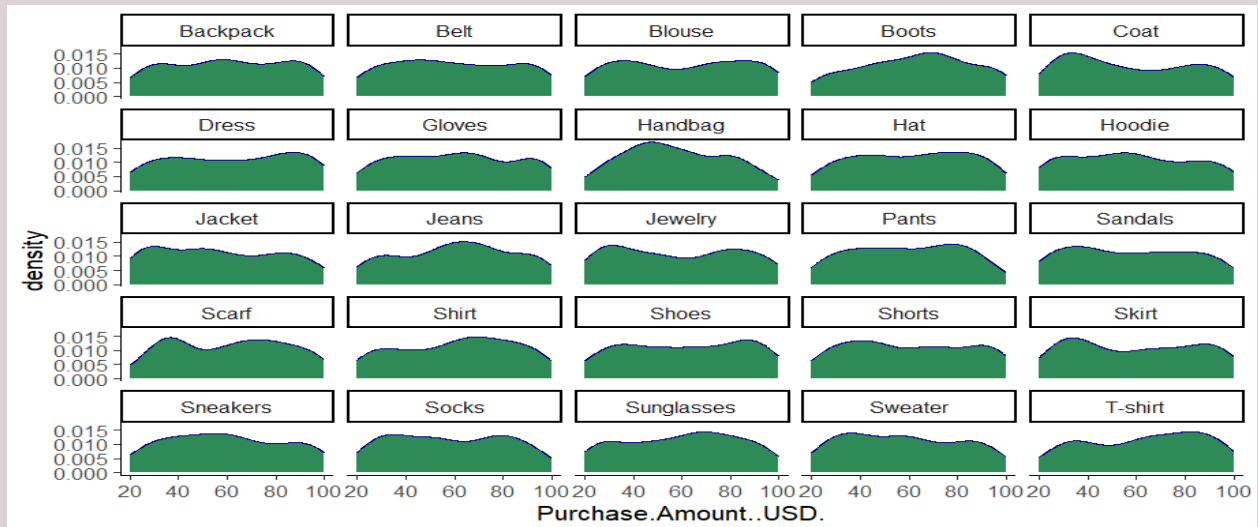


Figure 28. Density Plot of Purchases Across Different Items

2.8 DATA TRANSFORMATION

I came across an interesting variable that I wanted to visualize, but it was categorical. To proceed, I converted it into a numerical variable. I aimed to visualize the distribution of purchase amounts across different discount levels and explore the relationship between purchase amounts and the discounts applied.

To convert it into a numerical variable, here's how I made the changes.

```
> my_data <- my_data %>%
+   mutate(Discount.Applied = ifelse(Discount.Applied == "Yes", 1, 0))
```

R project - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

my_data my_data[is.na(Color),]

on	Size	Color	Season	Review.Rating	Subscription.Status	Payment.Method	Shipping.Type	Discount.Applied	Promo.Code.Used	Previous.Purchases	Preferred.Payment.Method	Frequency.of.Purchases
1	L	Silver	Summer	3.3	No	PayPal	Next Day Air	0	No	41	Debit Card	Weekly
2	S	Magenta	Summer	4.8	No	PayPal	2-Day Shipping	0	No	30	Venmo	Fortnightly
3	S	Black	Fall	4.5	No	PayPal	Next Day Air	0	No	24	Debit Card	Bi-Weekly
4	M	Olive	Summer	3.4	No	Credit Card	Store Pickup	0	No	6	Credit Card	Annually
5	L	Green	Winter	3.7	No	Cash	2-Day Shipping	0	No	14	Credit Card	Bi-Weekly
6	M	Yellow	Fall	4.9	No	Venmo	2-Day Shipping	0	No	37	Credit Card	Bi-Weekly
7	S	Lavender	Spring	3.5	No	Debit Card	Next Day Air	0	No	37	Venmo	Quarterly
8	M	Indigo	Spring	4.6	No	Cash	Standard	0	No	28	PayPal	Weekly
9	M	Pink	Winter	4.6	No	Venmo	Standard	0	No	43	PayPal	Bi-Weekly
10	L	Red	Fall	4.2	No	Credit Card	Store Pickup	0	No	25	Cash	Annually
11	S	Magenta	Winter	3.7	No	Venmo	Free Shipping	0	No	4	PayPal	Annually
12	S	Peach	Winter	3.0	No	Credit Card	Next Day Air	0	No	13	Venmo	Fortnightly
13	L	Turquoise	Winter	4.6	No	Credit Card	Store Pickup	0	No	32	Bank Transfer	Monthly
14	M	Black	Summer	3.2	No	Cash	Free Shipping	0	No	1	Cash	Annually
15	L	Blue	Summer	3.5	No	Credit Card	Next Day Air	0	No	4	Cash	Annually
16	M	Beige	Winter	2.6	No	Debit Card	Standard	0	No	33	Cash	Bi-Weekly
17	XL	Olive	Fall	4.9	No	Venmo	Standard	0	No	45	Bank Transfer	Fortnightly
18	M	Turquoise	Summer	2.7	No	Credit Card	Express	0	No	20	Debit Card	Fortnightly
19	S	Yellow	Fall	4.7	No	Venmo	2-Day Shipping	0	No	1	PayPal	Fortnightly
20	S	Turquoise	Spring	4.0	No	Bank Transfer	Free Shipping	0	No	7	Credit Card	Bi-Weekly
21	L	Lavender	Fall	3.5	No	Venmo	Next Day Air	0	No	48	Credit Card	Every 3 Months
22	XL	Turquoise	Winter	4.8	No	Credit Card	Standard	0	No	46	Bank Transfer	Quarterly
23	L	Teal	Spring	4.0	No	Bank Transfer	Free Shipping	0	No	34	Debit Card	Annually
24	L	Black	Fall	4.1	No	Credit Card	Express	0	No	36	Bank Transfer	Monthly

Showing 1 to 24 of 3,900 entries, 19 total columns

Console

Figure 29

2.8.1 HEAT MAP

Having successfully transformed the Discount Applied data into a numeric format, I will now analyze it alongside Purchase Amount to examine the density of the variables using the `stat_density` function.



Figure 30. Purchases Over Discount

(The heatmap shows the distribution of purchase amounts across discount levels, with color intensity indicating density. Purchases are most frequent at lower amounts with higher discounts, while higher amounts see fewer purchases. This suggests that larger discounts encourage more purchases, particularly for lower-priced items.)

2.9 CLUSTER ANALYSIS

Analyzing the four numerical variables on their own didn't provide enough insights, so I decided to combine them with other factors, such as customer location or product categories, for a deeper understanding. Variables like purchase amount or age should be more meaningful when analyzed alongside these categorical factors. However, since categorical variables cannot be directly used for cluster analysis, I first converted them into numerical values.

I then extracted the relevant categorical variables for clustering using the following command.

```
> Important_Categoricals <- data.frame(my_data[, c(5, 7, 10)])
```

I already installed the package, "fast Dummies" as I will convert the variables into numeric.

```
> library(fastDummies)
```

I converted the important_Categoricals data frame into numerical variables using the following command.

```
> Important_Categoricals <- dummy_cols(Important_Categoricals, remove_most_frequent_dummy = FALSE)
> View(Important_Categoricals)
>
```

I completed the dataset by merging the numerical variables with the transformed important categorical variables into one dataset for cluster analysis.

```
> variablesfor_kmeans <- cbind(my_data_numericals, Important_Categoricals[, 4:10])
> head(variablesfor_kmeans)
```

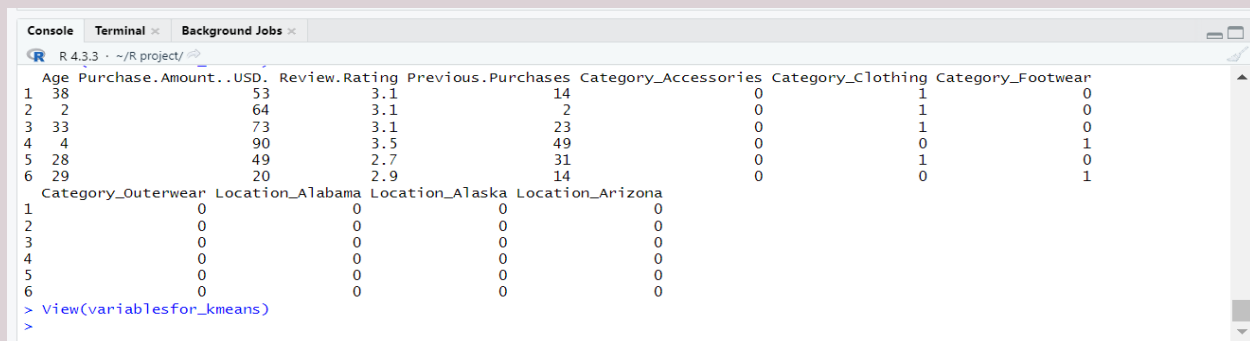


Figure 31

I also verified that the clustered data maintained its data.frame class.

```
> class(variablesfor_kmeans)
[1] "data. frame"
```

The final step before performing cluster analysis is determining the optimal number of clusters. I used the elbow method for this purpose. To proceed, I installed the “factoextra” package and utilized it to identify the appropriate number of clusters.

```
> library(factoextra)
> fviz_nbclust(variablesfor_kmeans, kmeans, method = "wss" )+
+   labs(subtitle = "Elbow Method")
```

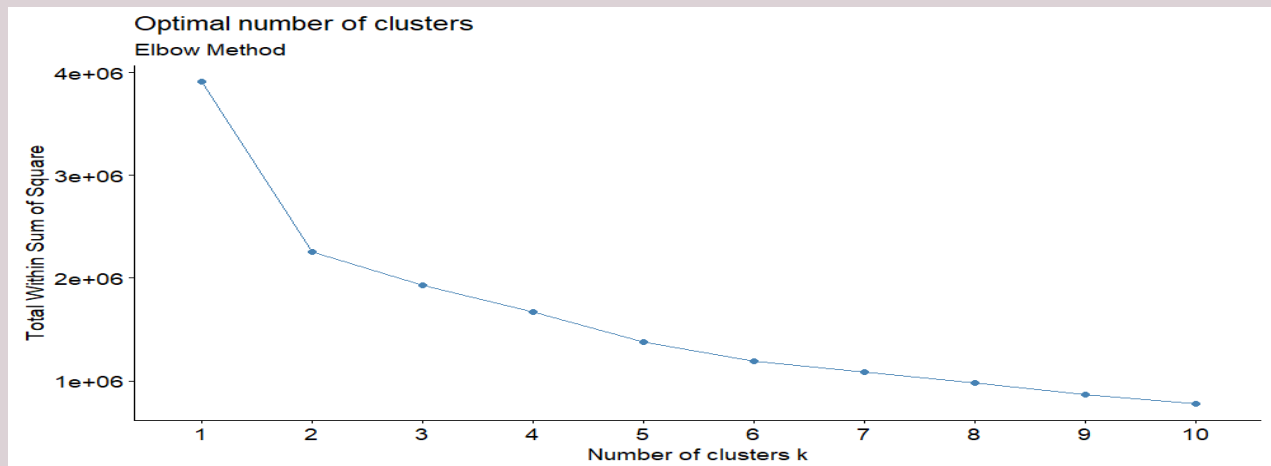


Figure 32. Cluster Chart

(The elbow plot shows how the total within-cluster sum of squares (WSS) decreases as the number of clusters increases. The decline slows after a certain point, creating an "elbow" shape. In this case, the elbow is around 3 clusters, suggesting this as the optimal number. Adding more clusters beyond this point provides minimal improvement in reducing WSS.)

I executed the following command to perform the cluster analysis.

```
> clusters <- kmeans (variablesfor_kmeans, centers = 8, iter.max = 10)
> clusters
```

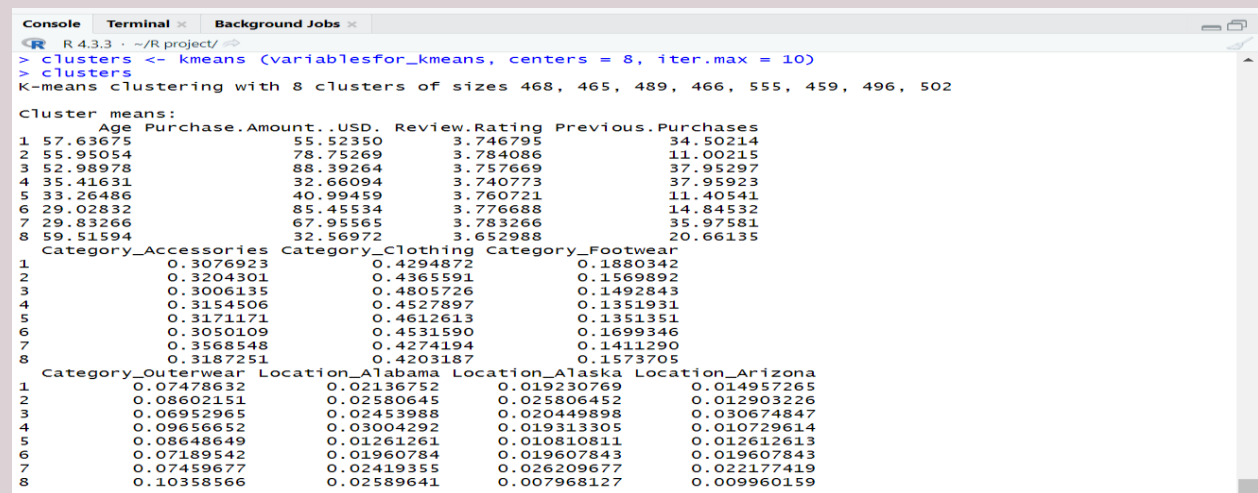


Figure 33

```

Console Terminal Background Jobs
R 4.3.3 ~ /R project/
Clustering vector:
[1] 1 6 2 7 1 8 3 5 6 8 8 6 1 1 1 2 4 8 8 3 7 7 8 7 5 5 5 2 3 5 8 7
[33] 7 4 3 4 7 5 4 1 3 1 6 6 7 8 1 1 1 8 4 2 4 4 3 4 7 7 1 2 8 2 4 7
[65] 2 1 6 4 8 2 4 4 5 3 2 2 1 7 3 3 7 3 5 4 4 2 4 1 4 2 4 6 2 7 3 3
[97] 7 6 7 4 3 7 1 2 8 2 8 7 3 3 8 6 7 5 2 3 2 8 1 7 3 6 4 3 3 1 4 6
[129] 5 7 5 4 7 1 3 2 6 1 8 3 8 1 7 5 5 7 8 1 3 6 6 5 7 1 6 2 4 6 7 8
[161] 4 1 3 2 2 5 4 5 6 5 4 5 1 4 4 2 3 4 8 8 2 5 6 8 4 4 7 6 5 6 1 3
[193] 8 6 2 8 3 1 3 7 8 8 8 6 1 5 2 8 3 5 6 1 1 6 6 3 8 5 7 4 2 8 4 4
[225] 2 5 6 3 6 8 3 5 8 6 5 3 7 3 8 1 1 3 3 6 1 1 6 6 3 8 5 7 4 2 8 4 4
[257] 8 5 2 5 5 6 5 4 6 5 3 1 8 8 5 1 4 8 8 2 5 4 3 7 2 7 6 5 8 5 5 8
[289] 7 3 4 8 3 8 8 8 6 8 8 6 6 3 4 7 3 1 5 5 8 8 3 1 6 3 5 8 4 3 2 2 2
[321] 1 4 3 3 7 1 1 1 5 8 1 3 6 2 4 7 6 5 2 5 4 4 1 4 7 1 5 4 1 6 7 6
[353] 6 1 3 5 3 5 4 7 8 3 2 3 2 3 8 8 4 4 8 7 6 5 3 8 5 8 1 4 7 8 7 2
[385] 7 4 3 5 7 4 2 7 6 1 1 1 6 7 3 8 5 1 1 2 1 8 2 5 3 8 2 3 3 2 2 3
[417] 7 7 7 2 5 4 1 8 8 6 2 2 6 6 4 7 3 3 1 4 4 1 8 1 6 3 7 7 4 5 6 2
[449] 4 4 5 8 4 1 6 3 3 3 3 8 1 6 3 7 7 3 4 7 5 4 2 3 5 4 2 7 6 1 4 3
[481] 3 2 5 6 1 1 7 7 8 3 1 2 2 5 7 5 7 4 3 6 4 8 8 8 4 2 4 8 4 6 6 8
[513] 5 7 1 6 5 1 6 5 2 4 3 5 1 4 1 4 3 6 8 1 7 1 2 8 2 5 4 1 4 4 7 2
[545] 7 4 3 5 7 4 2 7 6 1 1 1 6 7 3 8 5 1 1 2 1 8 2 5 3 8 2 3 3 2 2 3
[577] 2 7 2 1 2 6 2 7 2 8 8 7 2 2 7 8 5 8 8 6 7 4 2 7 8 7 5 5 6 5 7 4
[609] 4 1 2 5 1 1 2 3 7 2 2 5 6 4 1 7 2 2 6 5 3 6 3 1 5 1 7 7 5 4 4 4
[641] 3 2 8 5 3 8 1 2 4 3 5 4 8 7 7 5 5 2 8 8 3 1 6 2 3 7 4 7 3 3 1 4
[673] 1 7 4 2 7 1 7 3 8 7 4 5 3 1 8 4 7 6 7 8 3 1 4 6 4 1 1 5 2 2 6 2
[705] 7 7 1 8 5 2 6 2 3 7 7 5 6 4 8 7 1 4 7 7 1 6 1 6 7 8 4 7 8 1 1
[737] 7 8 8 8 7 6 8 7 1 2 4 2 5 2 6 6 1 5 8 3 5 1 5 8 3 4 8 8 2 8 6
[769] 3 2 5 7 4 6 5 5 2 8 5 3 4 5 5 1 7 1 8 3 1 6 5 1 4 5 8 4 3 3 4 1
[801] 4 5 5 4 1 4 1 6 3 3 3 4 1 8 5 6 4 4 3 8 8 8 7 8 3 6 3 7 7 5 2 1
[833] 3 2 2 5 6 4 4 8 8 3 3 2 1 4 3 1 3 7 8 6 7 1 5 2 4 7 4 2 6 3 6 4
[865] 2 8 2 4 6 7 2 4 8 2 6 5 5 1 7 2 7 1 4 4 3 3 7 1 3 3 1 4 7 5 7
[897] 8 7 2 1 3 1 5 3 2 5 4 6 1 8 8 7 4 7 4 8 2 8 6 7 8 2 8 3 6 2 6
[929] 8 3 6 5 8 3 5 8 6 5 6 6 5 4 7 5 3 5 5 8 2 4 8 1 7 1 2 2 3 5 1 6
[961] 5 1 7 6 5 5 1 5 1 4 2 2 1 3 2 8 3 4 8 8 3 4 7 4 6 6 3 1 3 8 1 6
[993] 3 3 2 2 6 8 3 8
[ reached getOption("max.print") -- omitted 2900 entries ]

```

Figure 34

```

Console Terminal Background Jobs
R 4.3.3 ~ /R project/
within cluster sum of squares by cluster:
[1] 106344.3 115179.7 118545.5 111755.9 144628.1 109605.3 120262.9
[8] 132347.2
( between_SS / total_SS = 75.5 %)

Available components:
[1] "cluster" "centers" "totss" "withinss"
[5] "tot.withinss" "betweenss" "size" "iter"
[9] "ifault"
>

```

Figure 35

Cluster Analysis Interpretation

Cluster 1:

This group consists of older customers who spend more, make frequent purchases, and primarily buy clothing.

Cluster 2:

Younger customers in this cluster spend less, have few prior purchases, and prefer accessories over other categories.

Cluster 3:

Middle-aged customers with moderate spending and balanced preferences for clothing and accessories. They tend to leave high review ratings.

Cluster 4:

Older customers with moderate spending and a specific focus on footwear.

Cluster 5:

This is a younger group that spends the least and has the fewest prior purchases, with no strong preference for any specific category.

Cluster 6:

Slightly older customers with moderate spending who mostly buy clothing and have a history of frequent purchases.

Cluster 7:

The youngest group in the analysis. They spend the least, have fewer prior purchases, and prefer accessories.

Cluster 8:

Middle-aged customers with higher spending show a strong preference for clothing and have moderate prior purchases.

Insights:

Customer Segments: Older customers with higher spending focus on clothing (e.g., Cluster 1), while younger, low-spending customers prefer accessories (e.g., Cluster 7).

Marketing Opportunities: Improve product quality or service for clusters with lower reviews (e.g., Clusters 2 and 5). Focus on upselling or premium offerings for high-spending clusters (e.g., Clusters 1 and 3).

3. SUMMARY AND CONCLUSION

I analyzed how people shop online. I looked at things like how much they spend, what they think of the products, how often they shop, and what kind of things they buy.

There is a slight connection between how much people spend and how good their reviews are. But it is not a strong connection. Similarly, people who shop more often tend to give slightly better reviews, but it is not a significant difference.

The cluster analysis provided deeper insights into customer segmentation, revealing distinct shopping patterns based on variables like age, gender, and purchasing behavior. Men tend to shop more than women, and clothes are the most popular item. People usually shop every 3 months, and they buy more things in the fall and winter. Medium-sized clothes are the most popular.

Discounts are important. People are more likely to buy things when they are on sale, especially for cheaper items. Diverse groups of people shop in diverse ways. By understanding these groups, businesses can better meet their customers' needs.

Overall, understanding how people shop online is important for businesses to succeed. By tailoring their strategies to diverse types of customers, businesses can improve their sales and customer satisfaction.